

Nathan Abel Ortiz

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EDUCATION

Northern Illinois University

B.S. in Mechatronics Engineering, Minor in Electrical Engineering, Minor in Mechanical Engineering

May, 2026

Dekalb, IL

SKILLS

- **CAD:** Fusion 360, SolidWorks
- **Programming:** MATLAB, Arduino, ROS, Python, Linux (Ubuntu)
- **Engineering:** PID control, SPICE, Computer Vision (OpenCV, YOLOv8), Inverse Kinematics
- **Fabrication:** 3D printing, laser cutters, CNC machines

PROJECTS

- **State competition 2nd place in skills USA Robotics: Urban Search and Rescue:**
Engineered a remote-controlled robotic system utilizing Arduino to navigate obstacle courses and perform payload retrieval, securing 2nd place at the SkillsUSA State Robotics Competition.
- **Obstacle course navigation robot**
Programmed an autonomous wheeled robot using Arduino and ultrasonic sensors to navigate through randomized obstacle layouts.
- **Mimic Glove (Real-time hand simulation)**
Developed a real-time motion capture system by integrating Arduino and flex sensors to accurately translate physical hand kinematics onto a 3D Unity model.
- **Indoor inspection robot**
Engineered a computer vision system mounted to a quadruped unitree GO2 utilizing ROS 2 and OpenCV to autonomously verify PPE compliance on in a warehouse/factory environment.
- **Rope climbing mechanism**
Designed and fabricated a mechanical rope-climbing system driven by a DC motor and a custom dual four-bar linkage mechanism.
- **Vision Tracking System with PD control**
Built a real-time facial tracking camera system using YOLOv8, implementing a custom PD control loop to optimize pan-tilt servo actuation and ensure smooth, continuous tracking.

WORK EXPERIENCE

Baxter Innovation Lab CLC

Lab Assistant

2021 – 2022

Grayslake, IL

- Maintained and troubleshoot lab prototyping equipment including 3D printers, laser cutters to support student and lab projects.
- Assisted with fabrication requests and helped ensure equipment was used safely and correctly.
- Diagnosed minor equipment issues, performed routine upkeep, and escalated larger repairs when needed.
- Updated and simplified machinery guides to improve usability for students and staff.

FedEx Ground

Package Handler

2019 – Present

Grayslake, IL

- Safely and effectively transfer packages through a warehouse facility and unload them at designated points.
- Adapt quickly to changing daily operational needs by providing cross-functional support across various warehouse work areas.